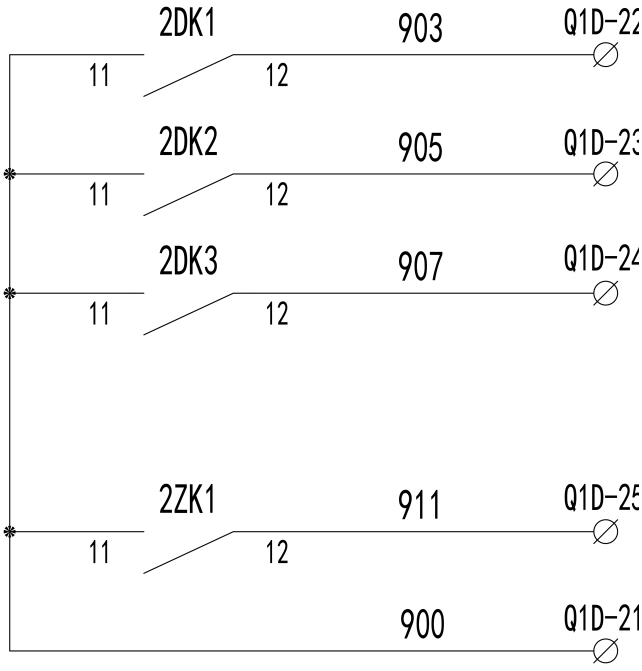


(a) Minimum contact closure

Question

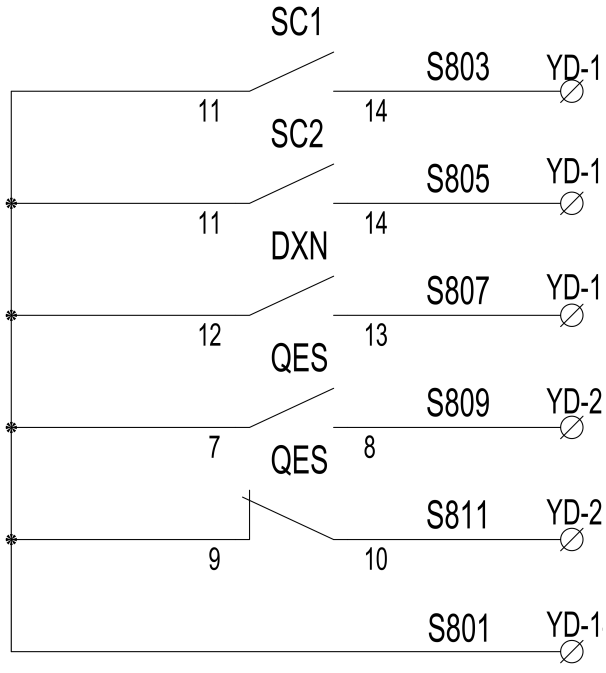
Which contacts must close to get continuity from 2DK1:12 to 2ZK1:12? Give every combination that works without extra contacts; leave the others open. Choose from 2DK1 contact 11–12, 2DK2 contact 11–12, 2DK3 contact 11–12 and 2ZK1 contact 11–12. Treat unlisted contacts as open.

Answer

2DK1 contact 11–12 and 2ZK1 contact 11–12 must be closed.

Evidence summary

2DK1:12 → 2DK1:11 → 2ZK1:11 → 2ZK1:12 requires both 2DK1 11–12 and 2ZK1 11–12. Opening either selected contact breaks continuity; no other minimal closure set works.



(b) Wire-only terminal groups

Question

Which of these terminals share a wire-only connection: DXN:12, DXN:13, QES:10, QES:7, QES:8, QES:9, SC1:11, SC1:14? Keep contact gaps separate and include isolated terminals.

Answer

{DXN:12, QES:7, QES:9, SC1:11}; {DXN:13}; {QES:10}; {QES:8}; {SC1:14}.

Evidence summary

External wires join DXN:12, QES:7, QES:9 and SC1:11. The other listed terminals form separate wire-only groups; contact gaps do not merge these groups.